Week 1 Lec 2: First Example of Data to Decision

From Infection Prediction to Health Care Policies

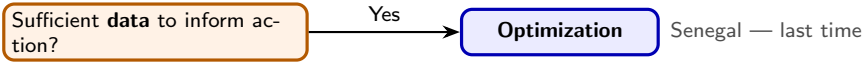
Peter Zhang

12768/19867/94867: From Data to Action

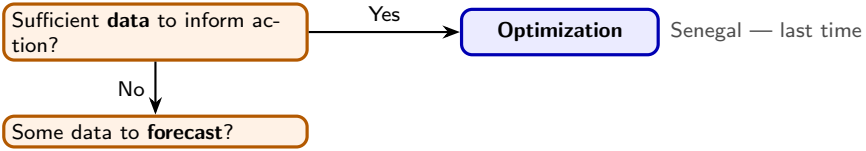
Where Today Sits: Choosing a Decision Framework

Sufficient **data** to inform action?

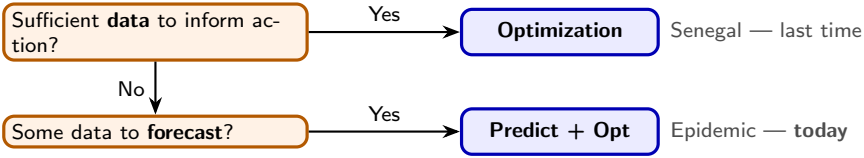
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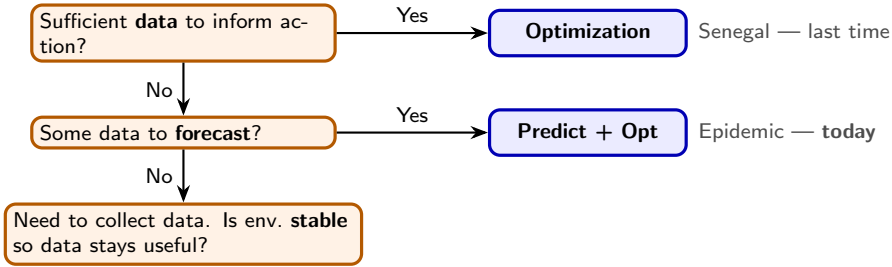
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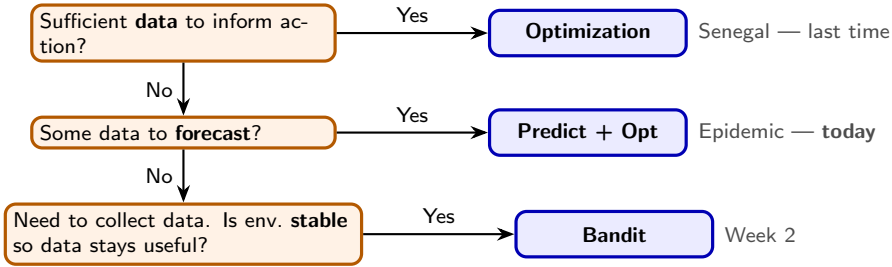
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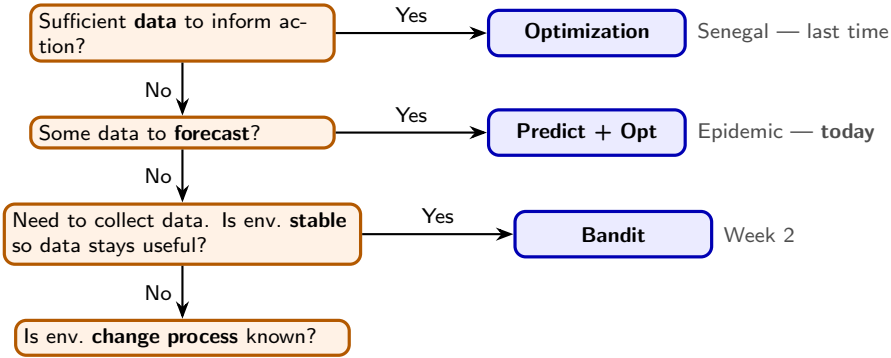
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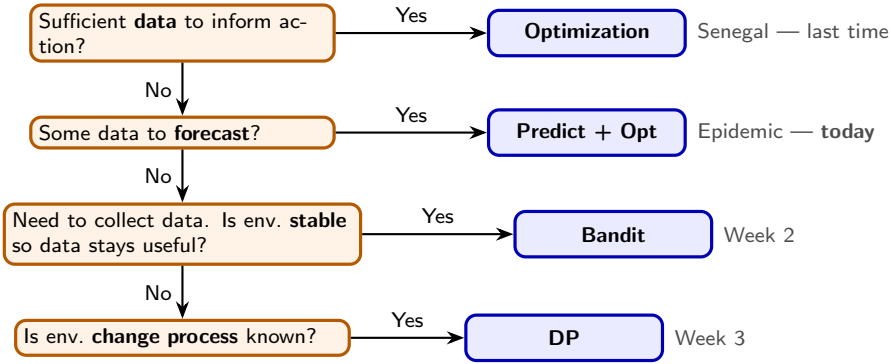
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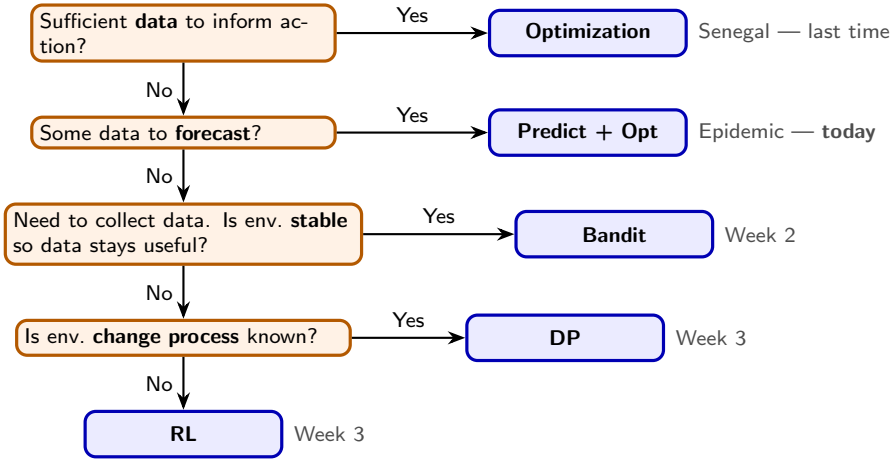
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- This lecture: from data to forecast to decisions
- (infections $\rightarrow$ health care policies)

Beyond Linear Programming

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But the number a decision turns on sometimes does not exist yet:

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So: before we can optimize anything, we have to **infer** the data the decision needs.

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- The question is very hard. Reopen too early and you can overwhelm hospitals that have very little slack to begin with. Reopen too late and the economic damage compounds, in places that can least absorb it.
- April, 2020: the Governor's office calls Ramayya Krishnan, Dean of Heinz College.

Carving Out a Question You Can Actually Answer

"Should we reopen the rural counties?" is not a question a model can answer.

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But we can break it down ... for instance:

- *If counties X, Y, and Z reopen, will their hospitals have enough regular beds, ICU beds, and ventilators?*

Krishnan's Ask

Ramayya Krishnan <rk2x@cmu.edu> Mon, Apr 27, 2020 at 11:26 PM
To: Hai Wang <haiwang@cmu.edu>, Peter Zhang <pyzhang@cmu.edu>
Cc: Rema Padman <rpadman@cmu.edu>

Dear Hai and Peter,

I received a call from the governors office today. Basically, the focus is on opening up a set a counties (rural counties) in PA. A part of this analysis is hospital capacity.

Here is how the model needs to be extended.

a. We need the inputs to the model to be differentiated by age. So doctors visits for people less than 45 and for people over 45.

b. We will need a set of parameters – transition probabilities that are dependent on age. If there is data from the literature that would be great. Duration can remain the same.

c. So in effect these two groups contribute in different proportions for the demand for hospital assets. However, they behave differently since older people have worse outcomes.

d. We would do the same monte carlo simulations.

e. We will want to use the data from DOH spreadsheet for total capacity and available capacity.

We need this working by the time we meet tomorrow. There is a meeting on Wednesday with a team from the governors office. Can you take a pass through this? I will have time in the afternoon for 30 mins or so but am pretty much back to back till the evening.

Thanks much for taking our ideas and making this an executable model that could potentially help with policy.

If you have questions about my email, we can set a time to exchange email or get on a quick call. Thanks.

Ramayya Krishnan

Dean, Heinz College of Information Systems and Public Policy

INFORMS 2019 President

The Data You Want vs. The Data You Have

What we wanted: how many people in these counties will need a bed, an ICU bed, or a ventilator over the coming weeks.

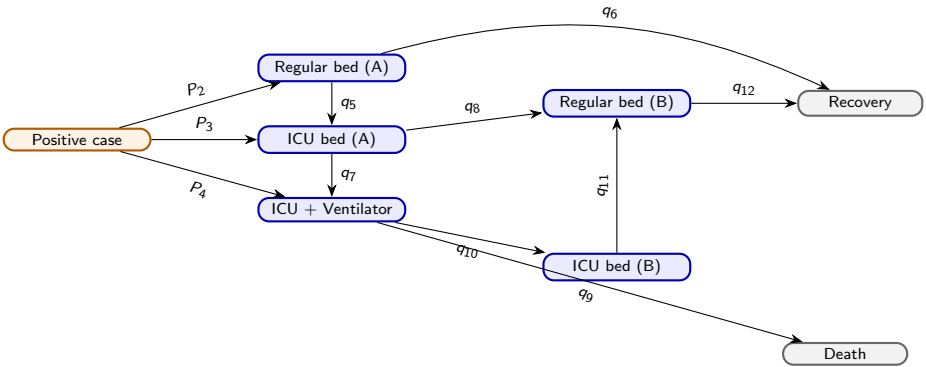
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What we wanted: how many people in these counties will need a bed, an ICU bed, or a ventilator over the coming weeks.
What existed: nothing like that. So we assembled substitutes.

- **Some data:** regional positive-case counts, split under-45 / 45-and-over, from Rayid Ghani – for three regions (North Central, North West, and the rest of the state). Effectively *one day's* worth, April 23.
- **The literature:** how a COVID patient actually progresses – how long in a regular bed, odds of escalating to ICU, odds of ventilation, how long on it – from a published ICU case series in the Seattle region (*NEJM*), estimated separately for the two age groups.
- **A structural model:** a map of the states a patient can be in, and the daily chance of moving between them.

Mechanism: How One Patient Moves Through the System

Every day, a Poisson number of new positive emerges, per region and age group. Most recover at home. The rest enter here (a simulated hospital environment):



Input and Output

In:

- Daily new positive cases, drawn as a **Poisson** random variable per region and age group, with the rate set from the April 23 counts
- Age-specific transition probabilities and durations from the literature
- PA Department of Health data on total and currently available capacity

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Sanity check (!): Hai compared our steady-state numbers against the IHME model's Pennsylvania estimates. Regular beds ≈1300 (ours) vs. 1400 (theirs); ICU 700 vs. 500; ventilators 600 vs. 400. Same ballpark, from a completely different method – that is what buys you the confidence to hand it over.

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- Thursday, April 30, from Krishnan: *"I included a slide for north central and northwest in the briefing for the governor which went well."*
- It was probably not the best prediction model anyone could build. It was the best model available **by Wednesday**.
- Two elements of the prediction model we aimed for: **interpretable**, and **comparable**.

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So we need another prediction model, of how an infection moves through a population.

Why Simple Forecasts Fail

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- Need a "mechanistic" model: how does flu *actually spread*?

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You supply the structure, and in exchange you can extrapolate and intervene (and also reduce data requirement). Need literature review and the sanity checks.

The SIR Model: Compartments

- $S(t)$ = number of people who are **susceptible** at time t
- $I(t)$ = number of people who are **infected** at time t
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Idea: At each step, some people move from S to I, and some from I to R.

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- Multiplying:
  - $I(t) \cdot \beta$ = total number of infectious contacts in period t
  - $\frac{S(t)}{N}$ = fraction of those contacts that hit a susceptible
  - $\Rightarrow$ product gives expected new infections in period t
- Minus sign: these individuals leave S and move into I

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Discussion:

- How would **mask-wearing** affect β?
- What about **school closures or remote work**?
- Does **population density** raise or lower β?
- How might **vaccination** change β?
- Viral factors: some viruses are naturally more contagious than others.

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Interpretation: The infected population grows if inflows $>$ outflows, and shrinks otherwise.

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Discussion:

- How does **access to healthcare** influence γ?
- What about **availability of antivirals**?
- How might **hospital capacity constraints** change recovery times?
- Do **age and comorbidities** affect γ?
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γ reflects biology, treatment, and system capacity.

Equation for Recovereds

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- Every period, a fraction γ of infecteds recover
- Once recovered, individuals stay immune
- R only grows. It never decreases in this model. ($\leftarrow$ is this a good assumption?)

Two Key Parameters

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Together, they govern:

- Speed of epidemic growth
- Height and timing of the peak

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Goal: Match the *weekly* model-predicted infection counts to observed data

Toy Observations: Weekly Active Infections (8 Weeks)

Setup: $N = 1{,}250{,}000$, $I_0 = 500$, $R_0 = 0$, $S_0 = N - I_0 - R_0$.

(Values are illustrative for teaching how we estimate β and γ.)

Week w	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
$I_{\text{obs}}(w)$	500	740	1050	1400	1910	2600	3550	4300

Notation: $I_{\text{obs}}(w)$ is the active infected count measured once per week.

Model vs. Data: What Are We Matching?

Model side: Simulate SIR (units: *per week*) and sample weekly to get

$$I_{\text{model}}(w; \beta, \gamma \mid S_0, I_0, R_0, N), \quad w = 0, \ldots, 7.$$

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Goal: Choose (β, γ) so that $I_{\text{model}}(w)$ **tracks** $I_{\text{obs}}(w)$ across all 8 weeks.

Precise Estimation Problem

Loss (mean squared error over 8 weeks):

$$\mathcal{L}(\beta, \gamma) = \frac{1}{T} \sum_{w=0}^{T-1} \Big(I_{\text{model}}(w; \beta, \gamma \mid S_0, I_0, R_0, N) - I_{\text{obs}}(w) \Big)^2, \quad T = 8.$$

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"Training" via Optimization:

$$\min_{\beta, \gamma} \ \mathcal{L}(\beta, \gamma) \quad \text{s.t.} \quad \beta \geq 0, \quad 0 < \gamma \leq 1 \text{ (per week)},$$

$$S_0 + I_0 + R_0 = N, \quad \text{and } (S, I, R) \text{ follow SIR dynamics between weeks.}$$

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Interpretation: Adjust **weekly** transmission β and recovery γ so the simulated weekly curve best fits the 8 observed points. (may want to control for the variation of β and γ values across weeks)

Coding

Code: `https://colab.research.google.com/drive/1o2mQ7NhIqew4mDkSgTpN3OIA1cKQ87ew?usp=sharing`

Goal: Fit SIR parameters to **weekly** observed infections

- Define data: N, (S_0, I_0, R_0), weeks $w = 0..7$, and $I_{\text{obs}}(w)$
- Implement SIR trajectory based on the differential equation dynamics
- Search over grids of β and γ; compute MSE between $I_{\text{model}}(w)$ and $I_{\text{obs}}(w)$
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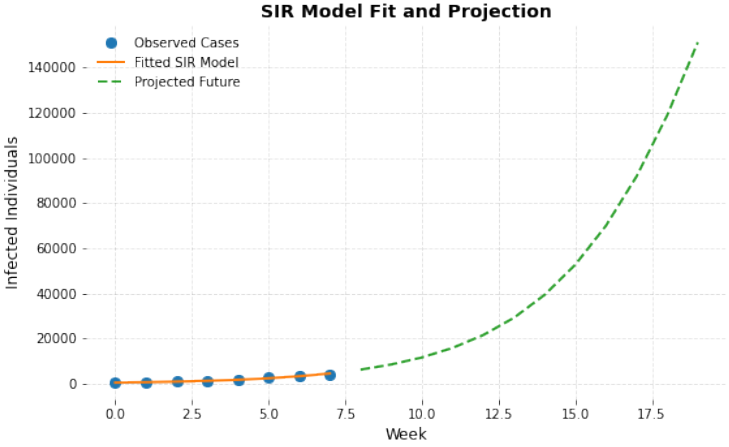
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Then visualize:

- Observed vs. fitted weekly $I(w)$ and a short **projection** beyond week 7

Visualization Example



A Year Later: The Same Shape of Question, in Jakarta

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Side note for your career: sometimes the bottleneck in a lot of public decisions is not policy authority or analytical skills alone. It is somebody in the room who can bridge both worlds.

July 2021: Delta Hits Indonesia

- Mid-July 2021: Indonesia's daily case count sets national records above 50,000 – that week, the WHO counted more new cases there than anywhere else in the world. Delta accounts for roughly 90% of Jakarta's cases.

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- So the live question, with a date attached: **is it safe to open, and if not yet, then when?** And does pushing vaccination faster change that answer?

We sent a memo on **July 25, 2021** – the day before that announced decision point.

Model What You Can Actually Decide

The policy question is about two levers: **how much to restrict mobility**, and **how fast to vaccinate**. Neither appears in the textbook SIR model – so put them in:

$$\frac{dS}{dt} = -\frac{\beta SI(1-q)}{N} - v_t v_e S, \qquad \frac{dI}{dt} = \frac{\beta SI(1-q)}{N} - \gamma I, \qquad \frac{dR}{dt} = \gamma I + v_t v_e S$$

- q = mobility restriction level: fewer contacts, so a smaller effective transmission term
- $v_t v_e$ = daily vaccination rate $\times$ vaccine effectiveness: a route out of Susceptible that does not pass through Infected

Model What You Can Actually Decide

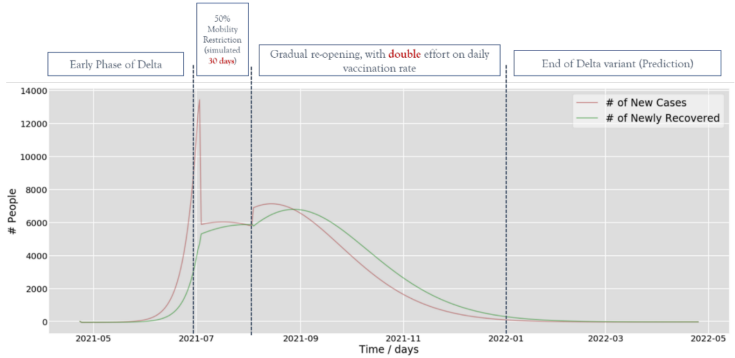
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This is the "L" in CLOUD. A model is only useful for a decision if the thing you can decide is a variable *inside* it.

The Recommendation



Hold 50% mobility restriction for 30 days, then relax gradually, while doubling the daily vaccination rate. Several scenarios were simulated (30 vs. 60 days of restriction, double vs. triple vaccination pace, Sinovac vs. AstraZeneca, and no reopening at all) so that the ministry could compare trade-offs rather than accept one number on faith.

How Did It Hold Up?

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How Did It Hold Up?

- **The policy:** emergency restrictions ran from July 3 and were held, in Level-4 form, into early August – then eased in stages. Close to the "hold about 30 days, then relax gradually" shape the memo recommended.
- **The epidemic:** by September 20, 2021, Indonesia reported 1,932 new cases – the lowest since August 2020, down roughly 94.5% from the mid-July peak.
- The model got the direction and the shape right – restrictions plus faster vaccination break the wave. It was **too pessimistic on timing**: it projected the wave fading through late 2021, and the collapse came faster than that.
- My suspicion was, the recommendation was conservative and safe; and it also probably aligned with the policymaker's intuition. TBH I do not know what would have happened if the policy we found was aggressive.

A Common Pattern

Different countries, different diseases, different decisions. Same four moves, because in both cases **the data the decision needed did not exist**:

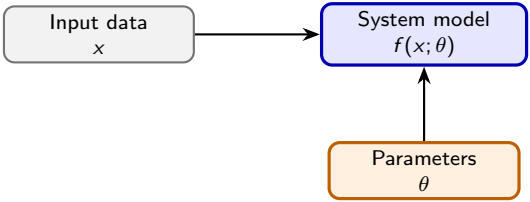
(a) **Build** a mechanism – states, transitions, rates – grounded in theory and literature

(b) **Fit** it to whatever real data you do have

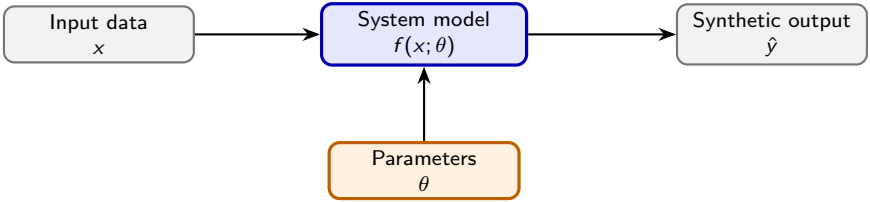
(c) **Predict** forward, past the last observation

(d) **Decide**, using that mechanistic prediction model

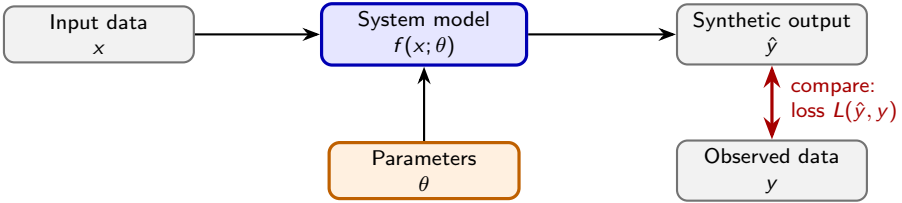
A Picture Representation



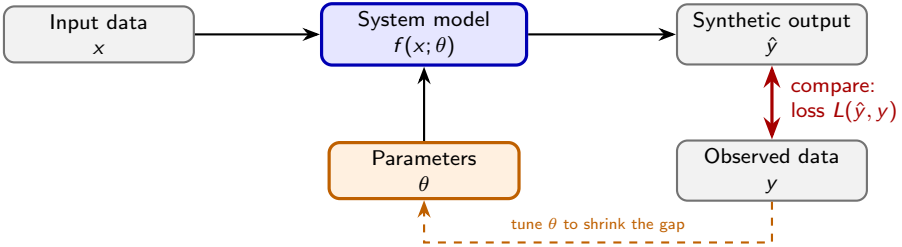
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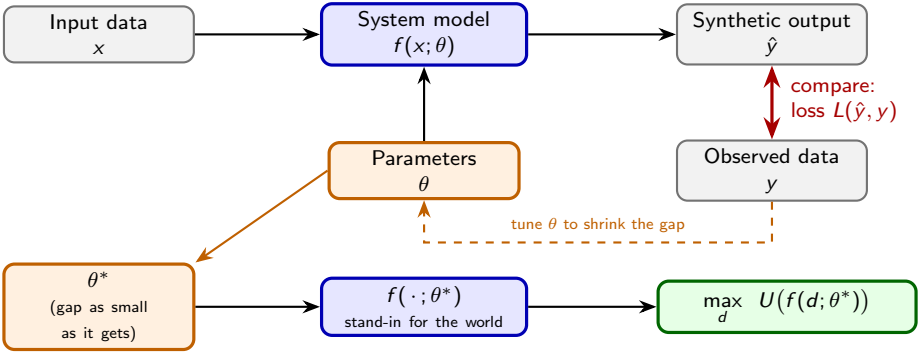
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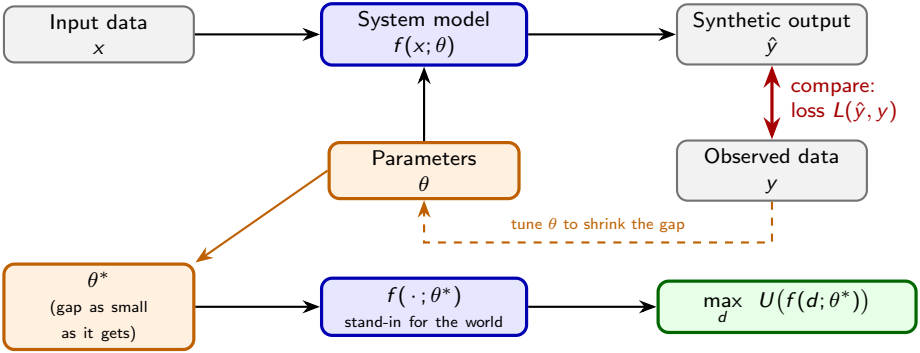
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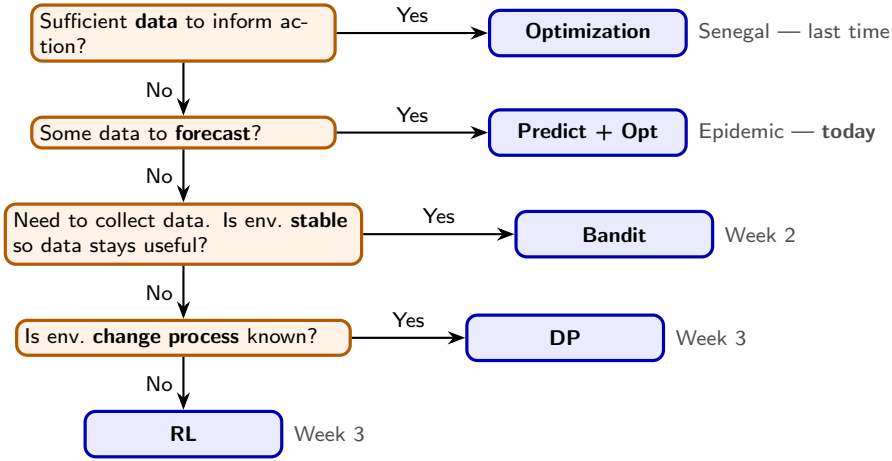


A Picture Representation



Two optimizations. The top loop searches over *parameters* to match reality for *prediction*. The bottom searches over *decisions* to get the best outcome.

Decision Framework



Appendix

From Flu Season to TikTok Season

New context: TikTok virality

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New context: TikTok virality

- Instead of patients sneezing in Pittsburgh...
- Think of teenagers scrolling in their bedrooms
- A sound or dance can "infect" the feed in hours
- Question: Can we use **SIR** to model TikTok trends?

Mapping SIR to TikTok Virality

- $S(t)$ = users who **haven't seen the trend** yet
- $I(t)$ = users who are **posting or duetting it**
- $R(t)$ = users who are **over it and moved on**

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Parameters:

- β = number of reposts/shares (catchiness factor)
- γ = probability a creator stops reposting (trend fatigue)

What Shapes β on TikTok?

- Is the sound catchy? ("Corn kid" vs. boring ad jingle)
- Algorithm boost: "For You Page" recommendations
- Influencers: Charli D'Amelio posts it $\rightarrow$ boom
- Shock value: weird, funny, or controversial content spreads faster

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If TikTok slowed resharing (like Twitter's "read before RT"), what happens to β?

What Shapes γ on TikTok?

- Trend fatigue: users move on after 3–5 days
- Competing memes: yesterday's dance is today's cringe
- Platform moderation: content takedowns, "shadow bans"
- Cultural cycles: some sounds burn bright, others simmer

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How could TikTok design features to raise γ for harmful content?

Decisions in the TikTok World

- Creators: How to ride the wave (maximize β)
- Platforms: When to intervene (fact-checks, downranking)
- Public health / info policy: How to "flatten the curve" of viral misinformation

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Think: TikTok dances and flu both spread fast. What tools do we have to slow one but not the other? What tools do we have to slow both?